

Component 1 : flood severity assessment and mapping

My component focuses on flood severity assessment and mapping. It uses rainfall, river water levels, DEM, historical flood records, land use, population, and satellite data to understand how severe a flood could be and which areas may be affected. The system estimates flood extent and potential impacts, then produces clear flood severity maps to support authorities in planning evacuations and disaster response.

Component 2 : Probabilistic Assessment and Forecasting of Landslide Occurrences using Machine Learning

Sri Lanka loses lives every monsoon season to landslides that could have been predicted. Our system changes that. We're building a machine learning-driven early warning engine that analyzes three key factors — historical landslide occurrences, rainfall patterns, and soil type — to calculate real-time landslide probability across high-risk areas. By learning from past landslide events and correlating them with live rainfall data and soil characteristics, the model identifies danger zones before they fail — not after. But prediction alone isn't enough. An Explainable AI layer, powered by SHAP, breaks down exactly which factor is driving each risk score, giving authorities a clear, transparent basis for every warning issued. The result is delivered through an inclusive, accessible Web-GIS live map — high-contrast themes for elderly citizens, and dual-language alerts in Sinhala and Tamil, bringing life-saving warnings straight to village level. From historical data to a warning a grandmother in Kotmale can understand — this is landslide prediction, made explainable, made local, made in time.

Component 3 : Human Impact and Decision Support

The third component is Human Impact and Decision Support. Flood and landslide models tell us how severe a hazard is — but not how many people it will actually affect. This component takes that severity, adds population and vulnerability data, and turns it into a clear decision — which area needs help first, and what action to take — so it can be passed on to the resource allocation team

Component 4 : The Emergency Response and Resource Coordination component

The Emergency Response and Resource Coordination component focuses on transforming high-priority disaster alerts into automated, real-time dispatch and evacuation actions for Sri Lanka's

disaster management authorities. It features an intelligent allocation framework that assesses live resource availability and prioritizes urgent districts during shortages, while generating safe, hazard-aware navigation routes that avoid active floods or landslides